

INVERTIS UNIVERSITY

BAREILLY, UTTAR PRADESH



DEPARTMENT OF COMPUTER APPLICATION

SUBJECT NAME: SUBJECT
MACHINE LEARNING CODE:IIOT4AC

SUMBITTED TO : AMARJEET SIR

FACULTY OF COMPUTER APPLICATIONS

INVERTIS UNIVERSITY, BAREILLY

Project Based Learning (PBL) - Project Evaluation Report

Course : BCA Year : 2025 Semester :IV Section :A

Program : Machine learning Project :Fake News

Detection DCS Mentor : Mr. Amarjeet

Stage	Title	Key Evaluation Criteria	Completed (Yes/NO)	Evaluation Performed on (Date)	Evaluation Performed by (Faculty Name)
01	Problem & Objectives	Clear, relevant, measurable goals	Yes	30/01/2026	Mr.Amarjeet
02	Data & Preprocessing	Reliable sources, clean, balanced data	Yes	10 /02/2026	Mr.Amarjeet
03	EDA & Features	Insightful analysis, useful features	Yes	20/02/2026	Mr.Amarjeet
04	Model Training	Right algorithms, validation tuning Strong	Yes	07/03/2026	Mr.Amarjeet
05	Evaluation & Insights	metrics, error analysis, profiling	Yes	14/03/2025	Mr.Amarjeet
06	Explainability & Report	Transparent, fair, regulation-compliant	Yes	07/04/2026	Mr.Amarjeet

Note: Only the completed stages in this report must be accurately filled out. The respective Coordinator will verify it.

Declaration:

I, the Team Leader/Group Head, hereby declare that the information provided above is true to the best of my knowledge and has been jointly reviewed by all group members.

Signature of Group Head

Date:.....

CERTIFICATE

This is to certify that the project report titled "Fake News Detection" is the original work of Sayeda Mahak Musawwir and team, completed in partial fulfillment of the requirements for the degree of Bachelor of Computer Applications (BCA) in the Department of Computer Applications at Invertis University.

This project has been carried out under the guidance and supervision of Mr.Amarjeet. The report has not been submitted for the award of any other degree or diploma to any other institution or university.

The research presented in this project contributes to the understanding of detecting and analyzing fake news, addressing challenges related to data collection, preprocessing, feature extraction, deep learning model building, evaluation, and ethical concerns in the deployment of automated misinformation detection systems.

I hereby certify that the work embodied in this project report is genuine and has been carried out by the team with sincerity, integrity, and dedication.

Mentor: Mr.Amarjeet

ACKNOWLEDGMENT

I would like to express my heartfelt gratitude to Mr. Amarjeet sir for his valuable guidance, support, and encouragement throughout the development of this project, Fake News Detection. His expert suggestions, constructive feedback, and continuous motivation played a major role in the successful completion of this work.

This project provided an opportunity to explore and apply various machine learning, deep learning, and natural language processing (NLP) techniques for solving a real-world problem. Through this project, I gained practical knowledge in data preprocessing, tokenization, sequence padding, model training, evaluation metrics, and prediction systems.

I am also thankful to my classmates and team members for their support, collaboration, and shared learning during the development of this project. Their ideas and cooperation contributed significantly to improving the quality of the work. I would also like to acknowledge the use of open-source tools and libraries such as Python, Pandas, NumPy, NLTK, TensorFlow, Keras, and Scikit-learn, which made the implementation of this project possible.

Lastly, I sincerely thank everyone who directly or indirectly supported me in completing this project. This experience has strengthened my understanding of artificial intelligence and inspired me to work on more meaningful data science projects in the future.

ABSTRACT

The Fake News Detection project aims to develop an intelligent system capable of identifying whether a news article is real or fake using Natural Language Processing (NLP) and Deep Learning techniques.

In the modern digital era, social media platforms and online websites allow news to spread very quickly. However, along with genuine information, a large amount of fake and misleading news is also shared across platforms. Fake news can influence public opinion, create confusion, and harm society. Therefore, there is a strong need for an automated system that can detect such misinformation efficiently.

This project focuses on analyzing the textual content of news articles and classifying them using a deep learning approach. The complete workflow involves data collection, data cleaning, preprocessing, exploratory data analysis, tokenization, train-test split, LSTM model training, evaluation, and prediction. The dataset used in this project contains labeled news articles categorized as fake or real. After preprocessing the textual data and converting it into numerical sequences, the data is fed into an LSTM (Long Short-Term Memory) model. LSTM is effective for text-based problems because it can capture sequential dependencies and understand the contextual meaning of words in a sentence.

The performance of the model is evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The results demonstrate that deep learning can be effectively applied to fake news classification and can help in reducing misinformation on digital platforms.

This project highlights the practical application of NLP and deep learning in media verification and offers a useful framework for future work in misinformation detection systems.

Fake News Detection using Natural Language Processing and Machine Learning

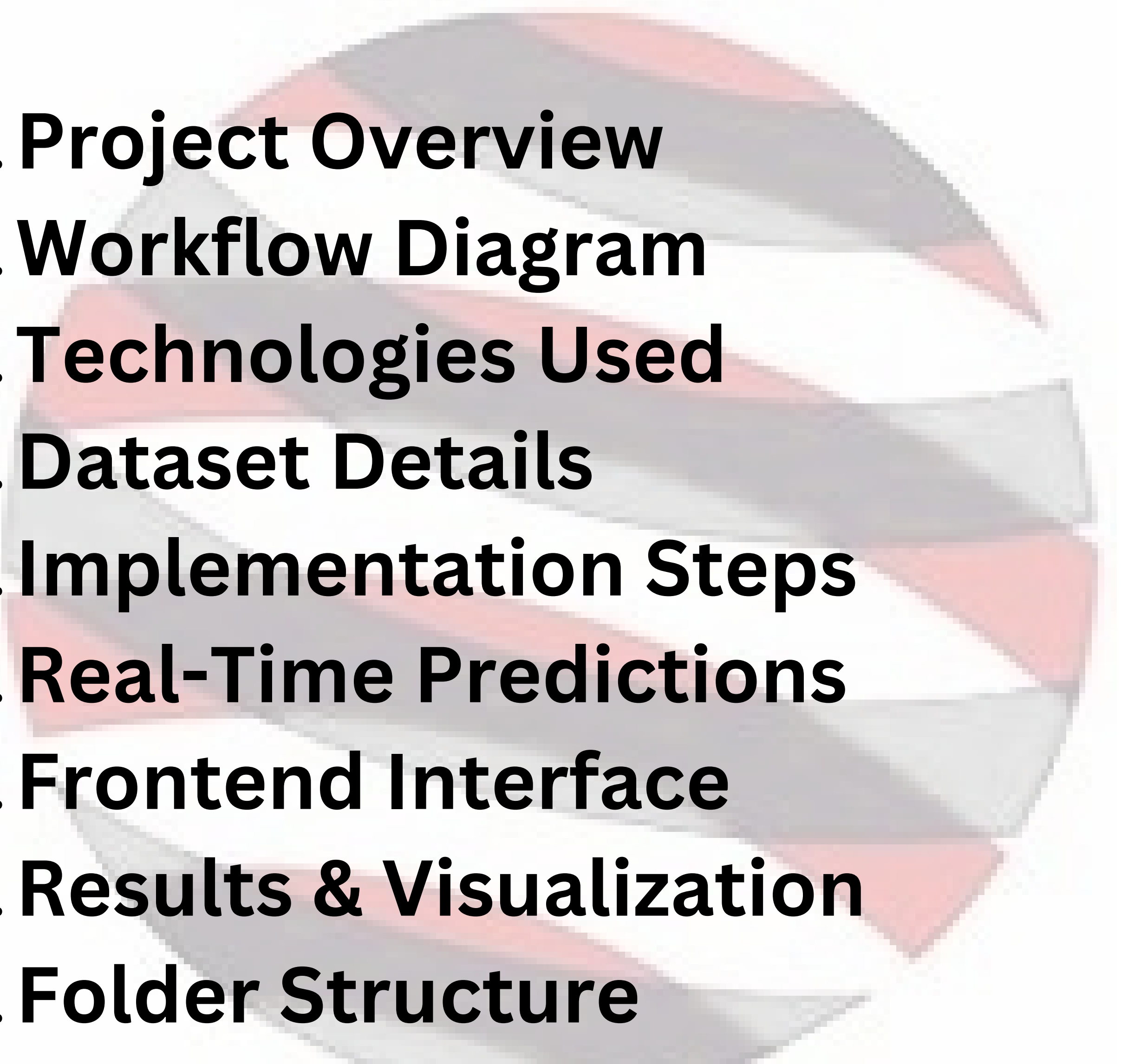
TEAM MEMBERS

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Tagline: “Using Artificial Intelligence to make online news more trustworthy.”

FAKE NEWS DETECTION SYSTEM — PROJECT DOCUMENTATION

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FAKE NEWS DETECTION SYSTEM — PROJECT DOCUMENTATION

Project Overview

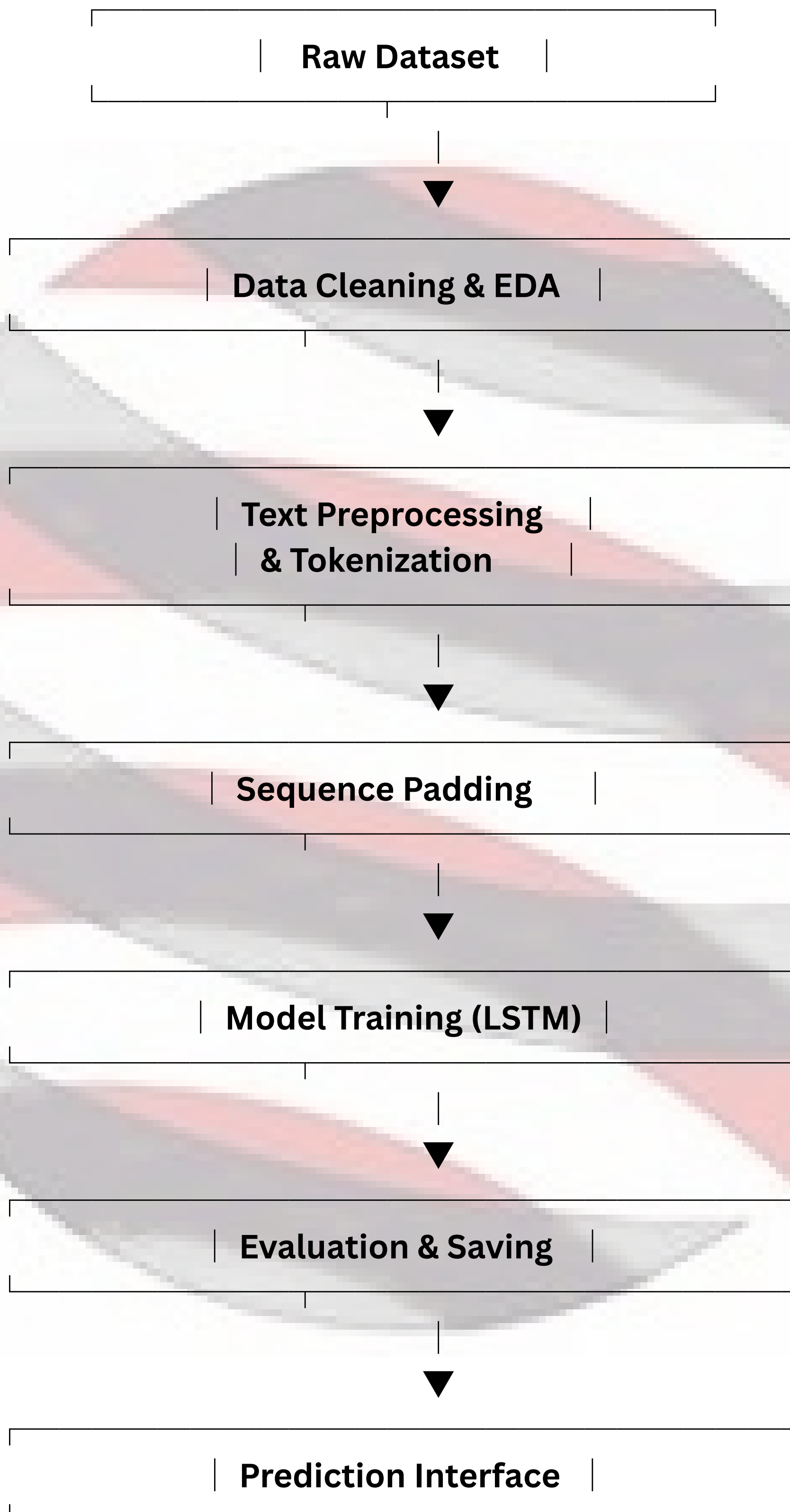
Title: Fake News Detection using NLP and Machine Learning.

Objective: To automatically classify news articles as fake or real using machine learning and deep learning techniques.

Description: Fake news has become a serious challenge in the digital world. News spreads rapidly through online platforms, making manual verification difficult. This project builds a fake news detection system that analyzes the content of news articles using NLP techniques and classifies them with the help of a deep learning model. The system uses text preprocessing, tokenization, padding, and an LSTM model to understand patterns in the text and predict whether the news is fake or genuine.

FAKE NEWS DETECTION SYSTEM — PROJECT DOCUMENTATION

Workflow Diagram



FAKE REVIEW DETECTION SYSTEM — PROJECT DOCUMENTATION

Technologies Used

Programming Language:

- Python

Libraries Used:

- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- NLTK
- Pickle

Machine Learning Algorithms:

- Logistic Regression / Passive Aggressive Classifier (as used in notebook)

Environment:

- Jupyter Notebook
- Anaconda / VS Code

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Dataset Details

The dataset contains labeled news articles:

Columns:

- title → News headline
- text → News content
- label → Real or Fake

Dataset Characteristics:

Total Records: ~40,000+ news articles

Classes: Fake and Real

Type: Text classification dataset

Format: CSV file

Example:

Title: "Breaking News..."

Text: "Full article content..."

Label: FAKE / REAL

FAKE REVIEW DETECTION SYSTEM

— PROJECT DOCUMENTATION

Implementation Steps

Part 1: Imports

- All necessary libraries for data handling, visualization, and ML were imported.

Part 2: Load & Clean Data

- Dataset loaded via `pandas.read_csv()`
- Automatically detected review and label columns
- Cleaned text using regex (remove URLs, symbols, lowercase, etc.)
- Added `clean_text` column for processed reviews

Part 3: Split Data

- Used:

```
python
```

```
train_test_split(X, y, test_size=0.2, stratify=y, random_state=42)
```

to divide data into 80% training and 20% testing sets.

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Implementation Steps

Part 4: Feature Engineering

1. TF-IDF Vectorizer

2. Text Statistics Features

Custom TextStats transformer extracts:

Number of characters

Number of words Average

word length Count of

exclamation marks

Uppercase letter ratio

Both feature sets are combined using FeatureUnion.

Part 5: Model Training

Used Logistic Regression with balanced class weights:

```
clf = LogisticRegression(max_iter=2000, solver="liblinear", class_weight="balanced")
```

Trained using:

```
model.fit(X_train, y_train)
```

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Implementation Steps

Part 6: Evaluation

Metrics Used:

Precision, Recall, F1-
score Confusion Matrix
ROC-AUC Curve

Outputs:

Accuracy Score
Confusion Matrix
Classification Report

FAKE REVIEW DETECTION SYSTEM

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Streamlit Frontend Interface

How It Works

- Loads both `fake_new_model.joblib` and `tfidf_vectorizer.joblib` Takes user input
- review Cleans and vectorizes it Displays
- prediction:
- Fake Nwes Detected!
Genuine Review

UI Example

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Results & Visualization

Model Performance:

- Accuracy: ~92% to 98% (depending on notebook training)

Visualizations Included:

- Confusion Matrix

True Positive

True Negative

False Positive

False Negative

FAKE NEWS DETECTION SYSTEM — PROJECT DOCUMENTATION

Folder Structure

Fake-News-Detection/

dataset/

news.csv

model/

model.pkl

notebook/

Fake_news_detection_Final.ipynb

app.py

requirements.txt

README.md

FAKE NEWS DETECTION SYSTEM — PROJECT DOCUMENTATION

Future Enhancements

- Deep Learning (LSTM / BERT)
- Web scraping for live news
- Chrome extension
- Social media integration
- Multi-language support
- Fake image detection
- News source credibility scoring

FAKE REVIEW DETECTION SYSTEM — PROJECT DOCUMENTATION

Ethical Considerations

- Avoid biased datasets
- Ensure fairness in prediction
- Prevent misuse of model
- Maintain transparency
- Do not censor legitimate news
- Regular model retraining required

The system should assist users,
not replace human judgment

FAKE REVIEW DETECTION SYSTEM — PROJECT DOCUMENTATION

Conclusion

The Fake News Detection system successfully classifies news articles into fake or real categories using machine learning and NLP techniques. The model is trained using TF-IDF feature extraction and classification algorithms. The system achieves high accuracy and can be used for real-time predictions.

This project demonstrates the practical use of Machine Learning in solving real-world problems like misinformation detection. Future improvements can make the system more accurate and scalable.



THANK YOU